

Learning = https://drive.google.com/drive/folders/1PwK_jWqfUtOjV7gHt8ODutq9QA5cxCGi

Lecture 1: Network Devices

A device can be both **client (service taker)** and **server (service provider)**. They are called end hosts / endpoints.

Switch provide connectivity within LAN.

Routers provide connectivity between LANs.

Network Firewall = Hardware can be placed inside or outside of network. They filter traffic based on rules implemented.

Host-based Firewall = Software. They filter traffic entering / exiting a host machine, like a PC.

Hub: all ports → one collision domain

Switch: each port → own collision domain, but also one broadcast domain

Router / VLAN: separate broadcast domains

Lecture 2: Interfaces and Cables

Ethernet Standards (copper)

Speed	Common Name	IEEE Standard	Informal Name	Maximum Length
10 Mbps	Ethernet	802.3i	10BASE-T	100 m
100 Mbps	Fast Ethernet	802.3u	100BASE-T	100 m
1 Gbps	Gigabit Ethernet	802.3ab	1000BASE-T	100 m
10 Gbps	10 Gig Ethernet	802.3an	10GBASE-T	100 m

UTP Cables

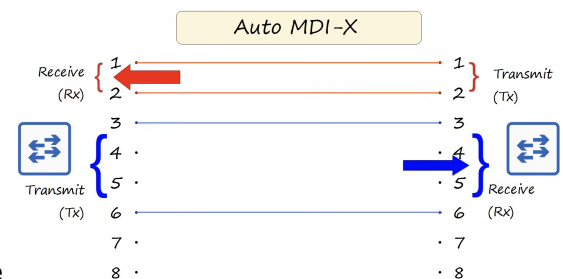
10BASE-T
100BASE-T = 2 pairs (4 wires)

1000BASE-T
10GBASE-t = 4 pairs (8 wires)

Full-Duplex: 2 Devices can send & receive to each other.

Device Type		Transmit (Tx) Pins	Receive (Rx) Pins
Router		1 and 2	3 and 6
Firewall		1 and 2	3 and 6
PC		1 and 2	3 and 6
Switch		3 and 6	1 and 2

Straight through cable, Crossover cable: based on



Auto MDI-X: Device changes its Tx and Rx based on communicating Device.

Fiber-Optic Cable Standards

Informal Name	IEEE Standard	Speed	Cable Type	Maximum Length
1000BASE-LX	802.3z	1 Gbps	Multimode or Single-Mode	550 m (MM) 5 km (SM)
10GBASE-SR	802.3ae	10 Gbps	Multimode	400 m
10GBASE-LR	802.3ae	10 Gbps	Single-Mode	10 km
10GBASE-ER	802.3ae	10 Gbps	Single-Mode	30 km

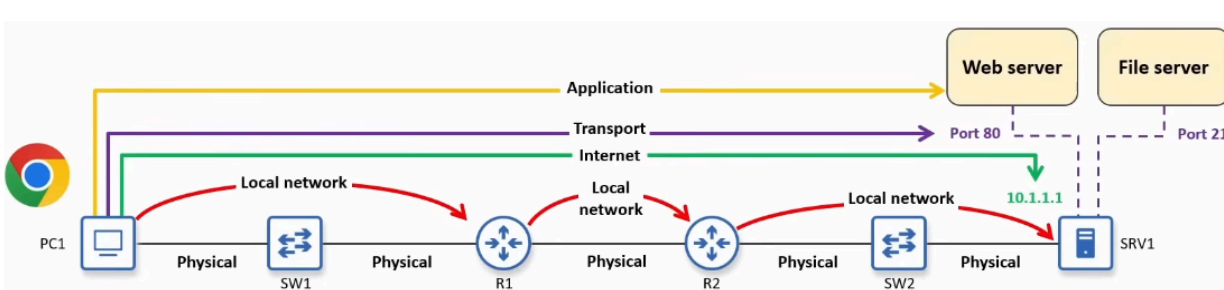
UTP

- Lower cost than fiber-optic.
- Shorter maximum distance than fiber-optic (~100m).
- Can be vulnerable to EMI (Electromagnetic Interference).
- RJ45 ports used with UTP are cheaper than SFP ports.
- Emit (leak) a faint signal outside of the cable, which can be copied (=security risk)

Fiber-Optic

- Higher cost than UTP.
- Longer maximum distance than UTP.
- No vulnerability to EMI.
- SFP ports are more expensive than RJ45 ports (single-mode is more expensive than multimode).
- Does not emit any signal outside of the cable (=no security risk).

Lecture 3: TCP/IP

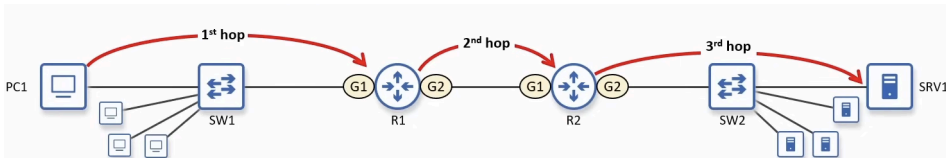


Common 5-layer model

Common 5-layer model	OSI model
7 Application	7 Application
4 Transport	6 Presentation
3 Network	5 Session
2 Data Link	4 Transport
1 Physical	3 Network
	2 Data Link
	1 Physical

Physical: Cables.

Data Link: LANs, Switches, MAC, Ethernet, WiFi. Switches do not count as one hop.



Uses **MAC (Media Access Control) addresses** to identify interfaces.

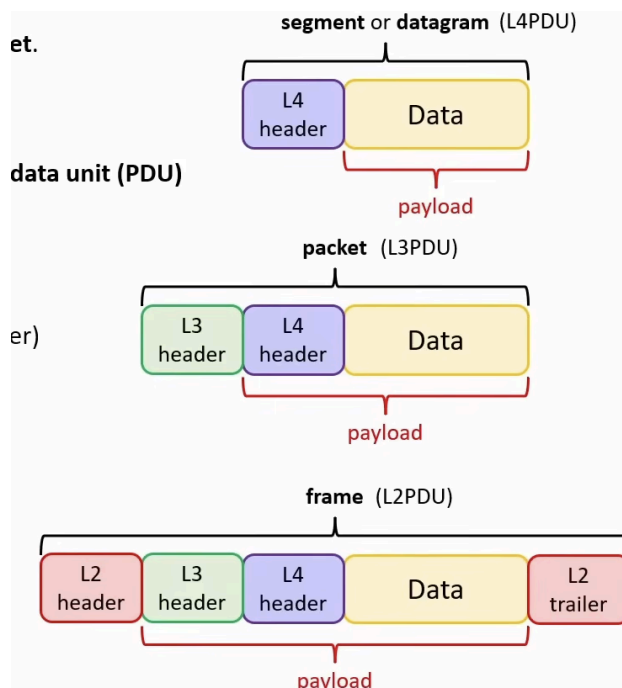
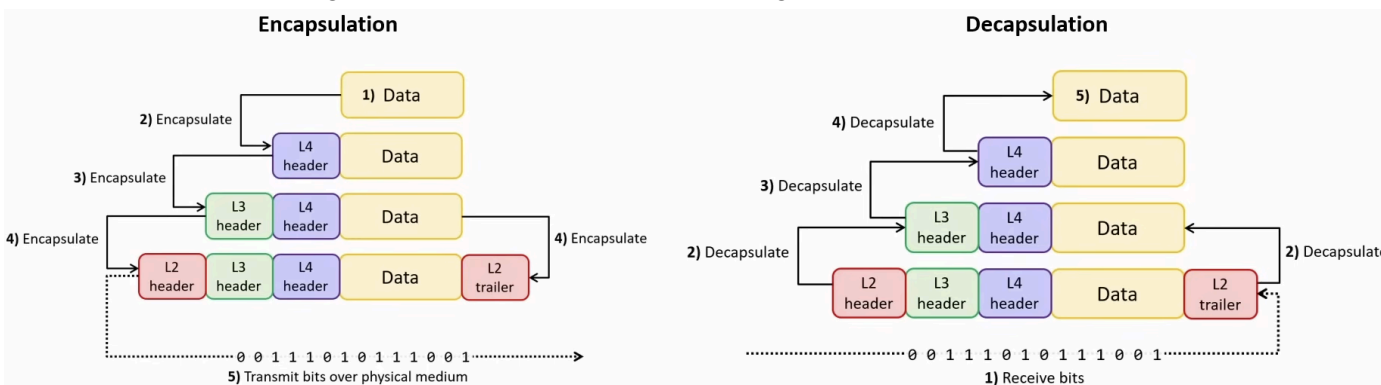
- PC1 sends the message to the MAC address of R1's G1 interface (NIC).
- R1 sends the message to the MAC address of R2's G1 interface (NIC).
- R2 sends the message to the MAC address of SRV1's interface (NIC).

Network: Routers, IP-Addresses, IP (v4, v6), ICMP (Internet Control Message Protocol).

Transport: Port Numbers, UDP/TCP.

Application: Browsing web servers: HTTP/S. Transferring Files: FTP, TFTP. Email...

Encapsulation = Sending Data. **Decapsulation** = Receiving Data.



TCP builds **segments**, UDP builds **datagrams**.

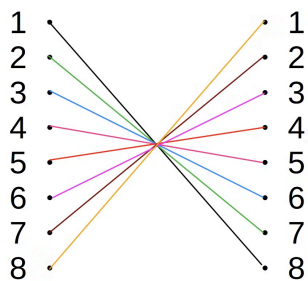
Adjacent-layer interaction: Each layer provide a service to its upper layer.

Same-layer interaction: Each layer communicates with the same layer on the other device.

Separation of layer: As long as each layer uses the same protocol with the same layer on the other device, different combinations are not a problem, such as: FTP, UDP or FTP, TCP.

Lecture 4: Intro to CLI

Rollover cable



Rollover cable is used to connect to RJ45 console port on a Cisco device

Configure the serial line	
Speed (baud)	9600
Data bits	8
Stop bits	1
Parity	None
Flow control	None

Connect the device to your computer. Use terminal Emulator (PuTTY) to access CLI. Select Serial.

> **user EXEC mode**

> **enable** – Enters **privileged EXEC mode**.

show startup-config – Displays the saved configuration that will be loaded after a restart.

show running-config – Displays the currently active configuration stored in RAM.

write / write memory / copy running-config startup-config – Copies current running configuration to startup configuration.

configure terminal – Enters **global configuration mode**.

(config)# **enable password <password>** – Configures a password for privileged EXEC mode. Password is case-sensitive.

(config)# **service password-encryption** – Encrypts plain-text passwords in the configuration.

(config)# **enable secret <password>** – Configures a secure, hashed password for privileged EXEC mode.

(config)# **do <privileged EXEC command>** – Executes a privileged EXEC command without leaving configuration mode.

(config)# **no <command>** – Removes or disables a configuration command.

? to view available commands.

*If you enable **service password-encryption**...*

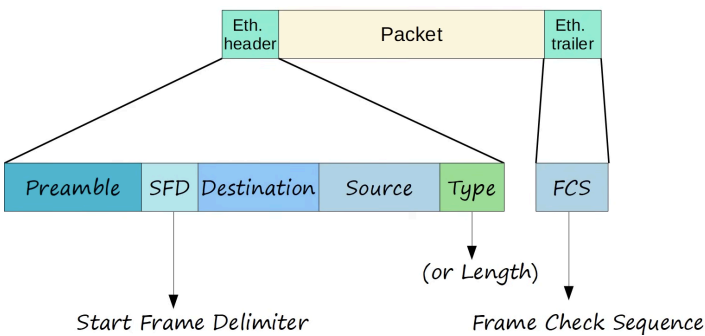
- current passwords *will be encrypted*.
- future passwords *will be encrypted*.
- the **enable secret** will not be effected.

*If you disable **service password-encryption**...*

- current passwords *will not be decrypted*.
- future passwords *will not be encrypted*.
- the **enable secret** will not be effected.

Lecture 5-6: Ethernet LAN Switching

LAN ends up on Router.



Preamble: 7 Bytes. Each byte = 10101010. Allows Devices to Synchronize their receiver clocks.

SFD: 1 Byte = 10101011. Marks end of preamble.

Destination & Source: 6 Bytes. MAC-Address = Burned-in-Address (BIA). Globally unique. First 3 bytes = OUI.

FCS: Length = 4 Bytes. 'Frame Check Sequence' Uses CRC-Algorithm to check data for errors.

Unicast-Frame = A frame destined for a single target. **Broadcast** for multiple targets.

How Switch works:

A switch builds a **MAC address table** ([show mac address-table](#)).

It reads the source MAC address and saves: **MAC address** → **port**.

If the destination MAC is unknown, the switch **floods** the frame to all ports except the incoming port = an **unknown unicast frame**. Devices that are not the destination drop/ignore the frame.

If the known unicast frame, the switch forwards the frame only to the correct port.

Dynamic MAC entries are removed after inactivity, usually after 5 minutes / 300 seconds.

Min payload (packet) size is 46 bytes. If less padding is added (0s).

ARP = Address Resolution Protocol. Learn MAC address of known IP address.

ARP Request (broadcast): send to all hosts on the network. FFFF.FFFF.FFFF is broadcast MAC address.

ARP Reply (unicast): sent only to the requester host.

ARP table: [arp -a](#) on host / [show arp](#) in cisco. IP Address (L3) → Physical / MAC address (L2).

Type static: default entry. Type dynamic: learned via ARP.

Ping: A network utility that is used to test reachability. Measures round-trip time. Uses two messages:

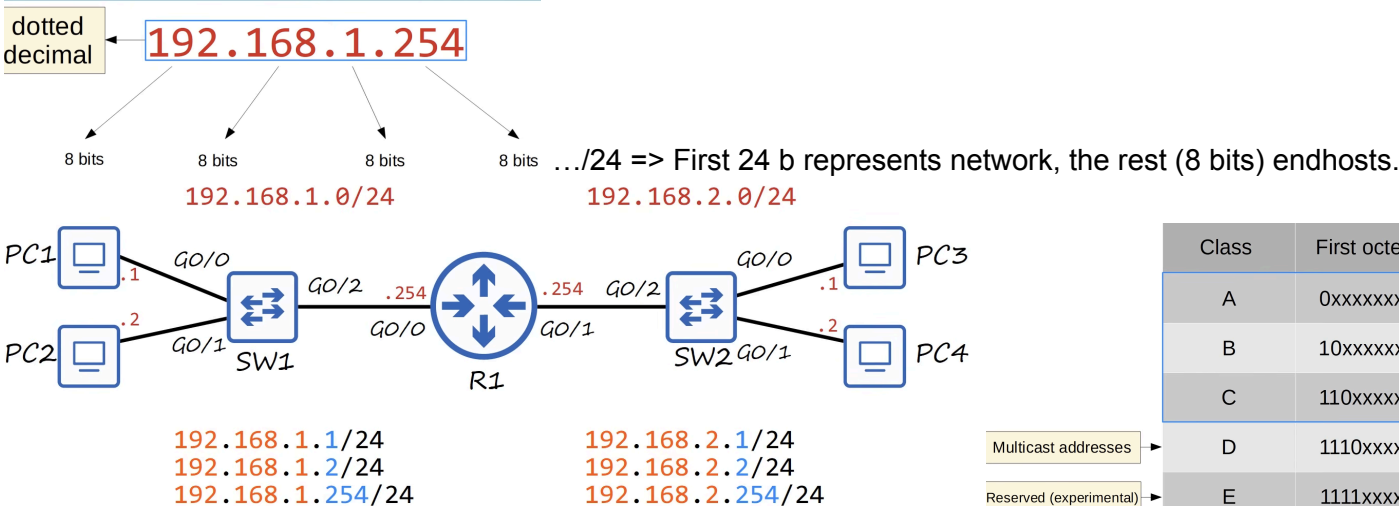
ICMP Echo Request / ICMP Echo Reply. Command to use ping: ping (ip-address)

Example: PC-1 sends ping to PC-3:

1. ARP Request (received by PC2, PC3, PC4)
2. ARP Reply (received by PC1)
3. ICMP Echo Request (received by PC3)
4. ICMP Echo Reply (received by PC1)

Lecture 7-8: IPv4 Addresses

IPv4 Addresses



Class	First octet	First octet numeric range
A	0xxxxxxx	0-127 ¹²⁶
B	10xxxxxx	128-191
C	110xxxxx	192-223
D	1110xxxx	224-239
E	1111xxxx	240-255

Multicast addresses →
Reserved (experimental) →

Loopback Addresses

- Address range 127.0.0.0 – 127.255.255.255
- Used to test the 'network stack' (think OSI, TCP/IP model) on the local device

```
C:\Users\user>ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\user>ping 127.23.68.241
Pinging 127.23.68.241 with 32 bytes of data:
Reply from 127.23.68.241: bytes=32 time<1ms TTL=128
Reply from 127.23.68.241: bytes=32 time<1ms TTL=128
Reply from 127.23.68.241: bytes=32 time<1ms TTL=128
Reply from 127.23.68.241: bytes=32 time<1ms TTL=128
Ping statistics for 127.23.68.241:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Class	Leading bits	Size of network number bit field	Size of rest bit field	Number of networks	Addresses per network
Class A	0	8	24	128 (2 ⁷)	16,777,216 (2 ²⁴)
Class B	10	16	16	16,384 (2 ¹⁴)	65,536 (2 ¹⁶)
Class C	110	24	8	2,097,152 (2 ²¹)	256 (2 ⁸)

Netmask

Class A: /8 255.0.0.0
(11111111 00000000 00000000 00000000)

Class B: /16 255.255.0.0
(11111111 11111111 00000000 00000000)

Class C: /24 255.255.255.0
(11111111 11111111 11111111 00000000) for Cisco devices.

Host portion full 0's = Network address.
Host portion full 1's = Broadcast address.
Both can not be assigned to an host.

Example: For a /24 network: Hostportion = 2⁸-2 = 254 Hosts possible.

IPv4 Addressing

```
R1>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES unset  administratively down down
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
GigabitEthernet0/3  unassigned      YES unset  administratively down down
```

- administratively down: Interface has been disabled with the 'shutdown' command.
- This is the default Status of Cisco router interfaces.
- Cisco switch interfaces are NOT administratively down by default.

show ip interface brief – Shows a brief overview of all interfaces, IP addresses, and status values.
show interfaces [interface] - Stats shows numbers of: Runts: Frames under 64 bytes. Giants: Frames over 1518 bytes.
CRC: CRC failed. Frame: Incorrect-formated frames. Input errors: Received errors. Output errors: Failed transmissions.
show interfaces description
clear mac address-table dynamic [interface interface-id] / [address mac-id]
(config)# interface GigabitEthernet0/1 – Enters interface configuration mode for GigabitEthernet0/1.
(config-if)# ip address 172.16.255.254 255.255.0.0 – Assigns the IP address 172.16.255.254/16.
(config-if)# no shutdown – activate the interface. shutdown – deactivate the interface.

Lecture 9: Switch Interfaces.

Routers: shutdown by default.

Switches: not shutdown by default. up/up if connected, down/down if not connected.

show interfaces status - trunk = Switch2Switch. duplex: full (send and receive) or half (send or receive) at the same time.

CSMA/CD: describes how devices avoid collisions in a half duplex situation. Process: Listen → Send → If (Collision):

Jamming Signal → Wait → Repeat.

(config-if)# **duplex <value>** - set duplex-value

(config-if)# **speed <value>** - set speed-value

(config-if)# **description <value>** - set description-value

(config)# **interface range f0/5 - 6, f0/9 - 12** - to config all of 5,6,9,10,11,12 interfaces at the same time. Called **interface**

range config mode.

Hub = Switch that floods everything.

if **autonegotiation** is enabled: every port uses full-duplex + max speed. If disabled: it uses slowest supported speed and if

its uses 10 / 100 Mbps => half-duplex. If it uses 1000 or more Mbps => full-duplex. It can cause Duplex-mismatch.

Lecture 10: IPv4-Header.

Version: Length = 4 bits. 0100 = IPv4, 0110 = IPv6.

IHL: Length = 4 bits. IPv4 header length. Value multiplied with 4 Bytes. Min: 5 (=20 Bytes), Max: 15 (= 60 Bytes).

DSCP: Length = 6 bits. Traffic priority.

ECN: Length = 2 bits. Network overload notification. 11 = overloaded → Receiver tells sender to slow→ Sender slower.3

Total Length: Length = 16 bits. Packet size in bytes. Value: 1 = 1 Byte. Min: 20 (=header without data) Max: 2^{16} .

Identification: Length = 16 bits. Used, if a packet is fragmented; same value on every fragment.

Fragment, if packet_len > MTU = usually 1500 bytes.

Flags: Length = 3 bits. Controls fragmentation: Bit 0: Reserved (always set to 0),

Bit 1: DF-bit == 1: Don't Fragment / 0 Fragment, Bit 2: MF-bit == 1 More Fragments / 0 Last Fragment.

Fragment Offset: Length = 13 bits. Identicates the position of the fragment, within the original unfragmented packet.

TTL: Length = 8 bits. Max #router-hops. A router will drop a packet if TTL == 0. Recommended = 64.

Protocol: Length = 8 bits. Protocol of L4PDU. Value 1 = ICMP, 6 = TCP, 17 = UDP, 89 = OSPF (dynamic routing protocol)

Header Checksum: Length = 16 bits. Checks the IPv4 header for errors.

Source Address: Length = 32 bits. Sender's IPv4 address.

Destination Address: Length = 32 bits. Receiver's IPv4 address.

Options: Variable length. Optional settings.

Padding: Variable length. Fills the header to a 32-bit boundary.

Lecture 11: Routing-Fundamentals.

Routing Methods (to learn routes): **Static** (manually configured) Routing / **Dynamic** (OSPF) Routing.

Routers store information about destinations in their **routing table**. Receive packets → table look up → forward.

If a router doesn't have a route that matches the received packet's destination, it will drop the packet.

If a router has multiple routes that match the received packet's destination, it uses the most specific matching route (matching longest prefix length).

Each route in the routing table is an instruction:

→ To reach destinations in network X, send the packet to next-hop Y (the next router in the path to the destination).

→ If the destination is a connected route (C), send the packet directly to the destination.

→ If the destination is your own IP address (local route (L)), receive the packet for yourself.

show ip route - view routing table on routers.

When an IP address on an interface configured and enabled, two routes are automatically added to the routing table:

Connected route (C): A route to the network connected to the interface. The network address calculated from the IP address and subnet mask.

Local route (L): A route to the exact IP address configured on the interface. The network address exact interface IP address with /32.

Example: Interface IP

192.168.1.1/24

192.168.1.130/25

10.20.30.45/16

Connected route

192.168.1.0/24 match 192.168.1.0-255

192.168.1.128/25 match 192.168.1.128-255

10.20.0.0/16 match 10.20.0.0-.255.255

Local route

192.168.1.1/32 match only 192.168.1.1

192.168.1.130/32

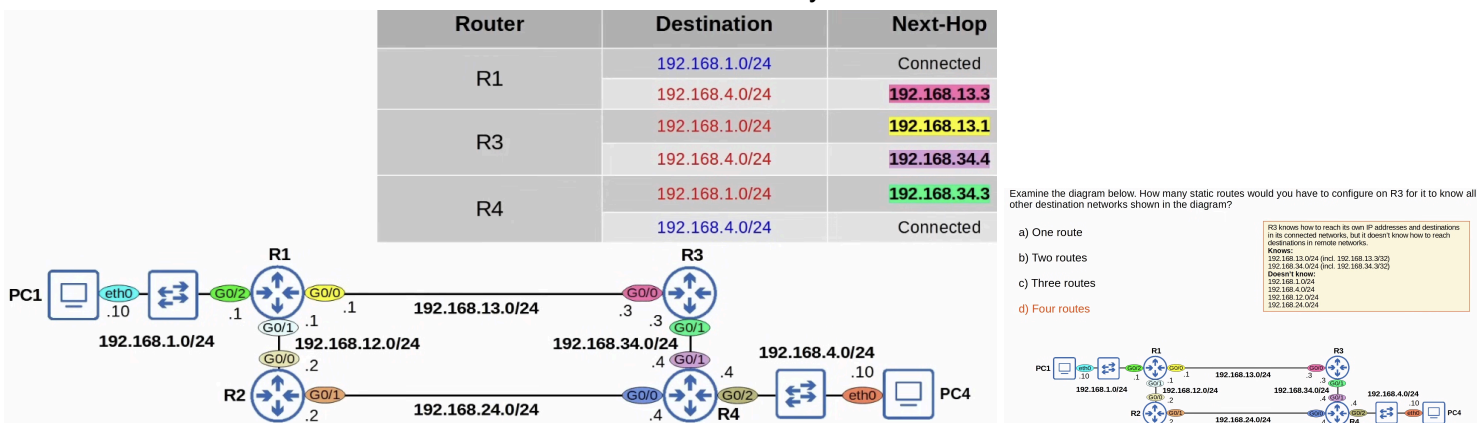
10.20.30.45/32

Static routes won't appear on the routing table automatically. Add them manually.

Gateway = Router. **Default route** = 0.0.0.0/0, if router wants to drop, it sends to this connection instead.

A host sends ARP, but the destination is out of the LAN, so the default router replies.

Traffic to destinations outside of the internal network is defaultly sent to the Internet.



Static Route Configuration:

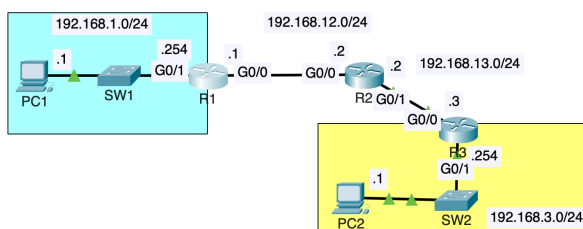
(config)# **ip route destination-ip netmask next-hop-ip**

(config)# **ip route destination-ip netmask exit-interface** (Avoid, since it is directly connecting => Proxy ARP)

(config)# **ip route destination-ip netmask exit-interface next-hop-ip**

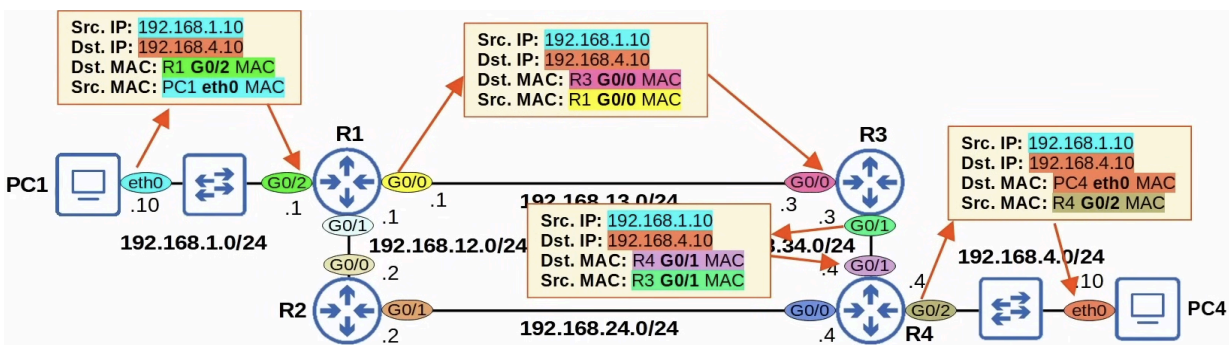
e.g.: R2(config)# **ip route 192.168.4.0 255.255.255.0 g0/0 192.168.24.4**

Learned from Labs: Configure first ip address of each interface and host. Then static route configuration.



R1(config)# **ip route 192.168.3.0 255.255.255.0 192.168.12.2**

Lecture 12: Life of a Packet.



Life of a Packet Through a Router

The router receives the Ethernet frame.

It removes the old Layer 2 header and trailer.

It checks the destination IP address.

It uses the routing table to select the next hop and outgoing interface.

If the next-hop MAC address is unknown, it sends an ARP request.

The router creates a new Layer 2 header and trailer.

The packet is forwarded to the next hop.

Src & Dst MAC Addresses change at every routed hop.

Src & Dst IP Address remain unchanged. (Exception: NAT changes them)

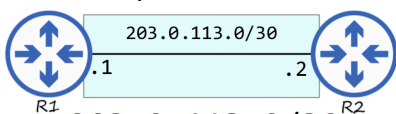
Lecture 13-15: Subnetting

IPv4 Address Classes

Class	First octet (binary)	First octet range (decimal)	
A	0xxxxxxx	0 - 127	0.0.0.0 ~ 127.255.255.255
B	10xxxxxx	128 - 191	128.0.0.0 ~ 191.255.255.255
C	110xxxxx	192 - 223	192.0.0.0 ~ 223.255.255.255
D	1110xxxx	224 - 239	224.0.0.0 ~ 239.255.255.255
E	1111xxxx	240 - 255	240.0.0.0 ~ 255.255.255.255

IANA assigns IP-Address to companies based on their size. **point-to-point-network** = Network between 2 routers.

CIDR (Classless Inter-Domain Routing): • With CIDR, the requirements of: Class A = /8, Class B = /16... were removed. So larger networks can be split into smaller networks. These smaller networks are called "subnetworks" or "subnets".



$203.0.113.0/30$
 $= 203.0.113.0 - 203.0.113.3$

1 1 0 0 1 0 1 1 . 0 0 0 0 0 0 0 0 . 0 1 1 1 0 0 0 1 . 0 0 0 0 0 0 0 0
 1 1 0 0 1 0 1 1 . 0 0 0 0 0 0 0 0 . 0 1 1 1 0 0 0 1 . 0 0 0 0 0 0 0 1
 1 1 0 0 1 0 1 1 . 0 0 0 0 0 0 0 0 . 0 1 1 1 0 0 0 1 . 0 0 0 0 0 0 1 0
 1 1 0 0 1 0 1 1 . 0 0 0 0 0 0 0 0 . 0 1 1 1 0 0 0 1 . 0 0 0 0 0 0 1 1

The remaining addresses in the 203.0.113.0/24 address block (203.0.113.4 - 203.0.113.255) are now available to be used in other subnets!

/31 also possible for point-to-point. BC and NA not used

Dotted Decimal	CIDR Notation
255.255.255.128	/25
255.255.255.192	/26
255.255.255.224	/27
255.255.255.240	/28
255.255.255.248	/29
255.255.255.252	/30
255.255.255.254	/31
255.255.255.255	/32

exercise: Divide the 192.168.255.0/24 network into five subnets of equal size.

Identify the five subnets.

Response: 1. Identify smallest x that fits into $2^x \geq \text{\#Subnets}$. Here $x = 3$. So we use /27.

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	0
NETWORK PORTION				HOST PORTION			

2. Take the value of the last bit of Network portion and calculate.

3. 192.168.255.0/27, 192.168.255.32/27, 192.168.255.64/27, 192.168.255.96/27, 192.168.255.128/27

exercise: What subnet does host 192.168.29.219/29 belong to? **Response:** 8 => 192.168.29.216/29

Process of subnetting Class A, B, C networks is the same!

Subnets/Hosts (Class C)

Prefix Length	Number of Subnets	Number of Hosts
/25	2	126
/26	4	62
/27	8	30
/28	16	14
/29	32	6
/30	64	2
/31	128	0 (2)
/32	256	0 (1)

Subnets/Hosts (Class B)

Prefix Length	Number of Subnets	Number of Hosts
/17	2	32766
/18	4	16382
/19	8	8190
/20	16	4094
/21	32	2044
/22	64	1022
/23	128	510
/24	256	254

Prefix Length	Number of Subnets	Number of Hosts
/25	512	126
/26	1024	62
/27	2048	30
/28	4096	14
/29	8192	6
/30	16384	2
/31	32768	0 (2)
/32	65536	0 (1)

exercise: What subnet does host 172.25.217.192/21 belong to? **Response:** 3 8 => 175.25.216.0/21

exercise: Divide the 172.16.0.0/16 network into 4 subnets of equal size. Identify the network and broadcast addresses.

Response: /18 => 64

S1: N = 172.16.0.0 B = 127.16.63.255

S2: N = 172.16.64.0 B = 172.16.127.255

S3: N = 172.16.128.0 B = 172.16.191.255

S4: N = 172.16.192.0 B = 172.16.255.255

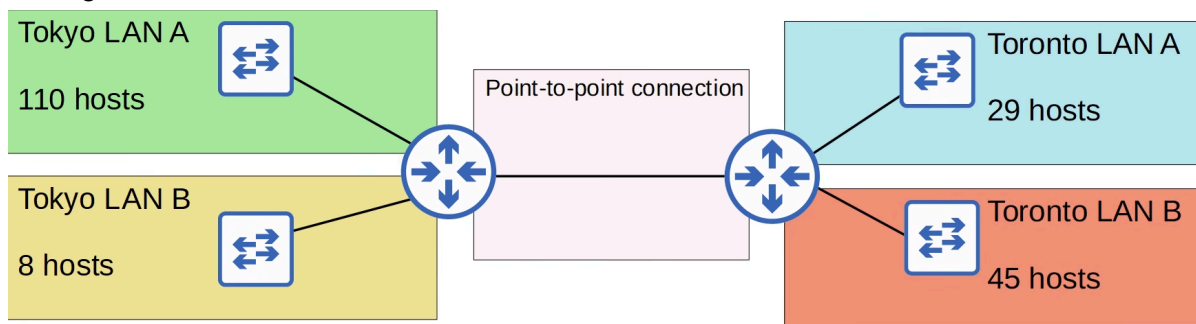
exercise: You have been given the 10.0.0.0/8 network. You must create 2000 subnets which will be distributed to various enterprises. What prefix length must you use? How many host addresses will be in each subnet?

Response: #Subnets: Identify smallest x that fits into $2^x \geq \text{\#Subnets}$. Here $x = 11$. So we use $8+11 = /19$.

#Host addresses: $32 - 19 = 13$. $2^{13} = 8192$.

FLSM (Fixed-Length Subnet Mask): is the process of creating subnets of the same sizes. Used till this point.

VLSM (Variable-Length Subnet Masks): is the process of creating subnets of different sizes. Assign firstly the largest then second-largest until shortest subnet.



192.168.1.0/24

exercise:

Tokyo Lan A: 192.168.1.0/25

Toronto Lan B: 192.168.1.128/26

Toronto Lan A: 192.168.1.192/27

Tokyo Lan B: 192.168.1.224/28

ptp LAN: 192.168.1.240/29

For more practice:

<http://www.subnettingquestions.com/>

<http://subnetting.org/>

<https://subnettingpractice.com/>

Learned from labs: Use last usable address to assign default gateway (routers interface).

Route Destination should be Network Address. 192.168.5.128/26 => **ip route 192.168.5.128 255.255.255.192 next-hop-ip**

Lecture 16-18: VLANs

VLAN = logically separates a Layer 2 network into different broadcast domains.

→ Broadcast traffic stays inside its VLAN.

→ Inter-VLAN communication → requires Layer 3 routing (Router / Layer 3 Switch).

Access Port

→ carries traffic for one VLAN

→ usually connects end devices

Trunk Port

→ carries traffic for multiple VLANs

→ uses IEEE 802.1Q VLAN tags

→ usually switch ↔ switch / switch ↔ router

2 trunking protocols: **ISL** (Inter-Switch Link) and **IEEE 802.1Q** (dot1q).

802.1Q tag = Ethernet VLAN tagging standard. VLAN tag contains **VID** (VLAN ID).

Usable VLAN IDs: 1–4094 (0 and 4095 reserved).

TPID (Tag Protocol ID) = 0x8100.

PCP (Priority Code Point): prioritizes important traffic in congested networks.

DEI (Drop Eligible Indicator): Indicates frames that can be dropped if network is congested.

802.1Q tag format

16 bits	3 bits	1 bit	12 bits
TPID	TCI		
	PCP	DEI	VID

Native VLAN → normally untagged traffic on trunk

ROAS (Router on a Stick) → one router interface + multiple VLAN subinterfaces

Alternative to ROAS: 1 connection for each VLAN between R and S.

#show vlan brief

Configure Switch:

(config-if-range)#**switchport mode access** - Configures the selected switch ports as access ports.

(config-if)#**switchport access vlan <vlan-id>** - Assigns an access port to a specific VLAN.

(config)#**vlan <vlan-id>** - new vlan

(config-vlan)#**name <name>** - name vlan

#show interfaces trunk - Displays the configured trunk ports and their VLAN information.

(config-if)#**switchport trunk encapsulation dot1q** - Configures the trunk port to use IEEE 802.1Q encapsulation.

(config-if-range)#**switchport mode trunk** - Configures the selected switch ports as trunk ports.

(config-if)#**switchport trunk allowed vlan <vlan-id>** - specifies which VLANs are allowed to pass through a trunk port.

(config-if)#**switchport trunk native vlan <vlan-id>** - Changes the native VLAN of the trunk port. To 1001.

Configure Router:

(config-if)#**interface interface-id <vlan-id> e.g.: interface g0/0.10** - Creates and enters a router subinterface.

(config-subif)#**encapsulation dot1q <vlan-id>** - Assigns the subinterface to a VLAN using 802.1Q encapsulation.

(config-subif)#**encapsulation dot1q <vlan-id> native** - Assigns native VLAN.

(config-subif)#**ip address ip netmask** - Assigns an IP address and subnet mask to subinterface. Address = default gw.

L3 (Multilayer) Switches: capable of both switching and routing. It is “Layer 3 aware”.

You can assign IP addresses to its interfaces, like a router.

You can create virtual interfaces for each VLAN and assign IP addresses to those interfaces.

You can configure routes on it, just like a router.

Preferred method for inter-VLAN routing.

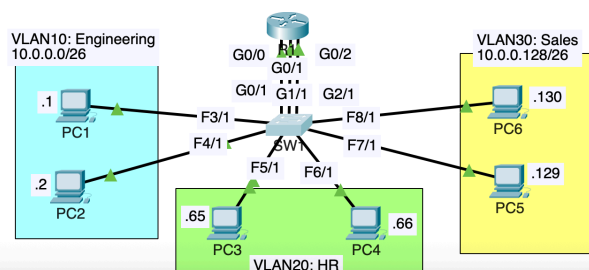
Inter-VLAN Routing via SVI (Switch Virtual Interfaces): An SVI is a virtual interface for a VLAN. Its IP address is used as the default gateway, and the switch routes traffic between VLANs. Shutdown by default.

(config)#**ip routing** - enables Layer 3 routing on the switch.

(config-if)#**no switchport** - configures the interface as 'routed' (L3 port) **interface vlan10** => assign Ip-Address => **no shutdo**

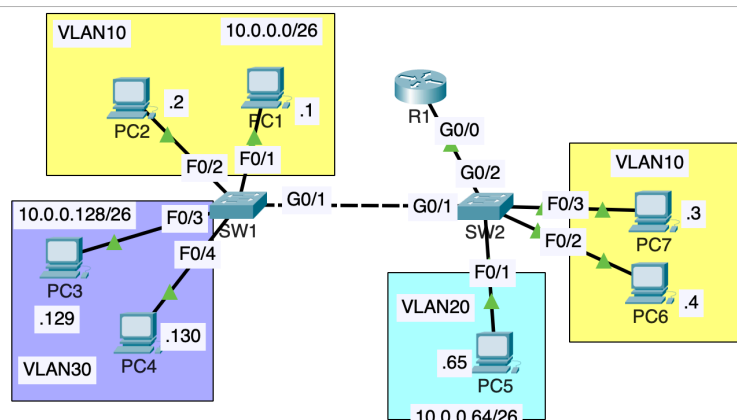
default interface g0/1 - remove configurations.

Learned from labs: A VLAN has: interfaces between hosts in them to switch and interface between switch and router.



VLAN	Name	Status	Ports
1	default	active	Fa9/1
10	Engineering	active	Gig0/1, Fa3/1, Fa4/1
20	HR	active	Gig1/1, Fa5/1, Fa6/1
30	Sales	active	Gig2/1, Fa7/1, Fa8/1

With multiple Connection:



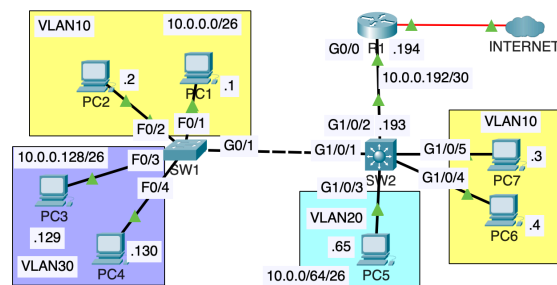
With ROAS:

SW1–SW2 trunk must allow VLAN 30 so VLAN 30 traffic from SW1 can reach its default gateway on R1 through SW2.

SW1–SW2 trunk does not allow VLAN 20 because no VLAN 20 traffic needs to cross the link between SW1 and SW2.

SW2 interface g0/1: vlan 30.

R1-SW2 trunk includes VLAN 10,20,30.



With L3-Switches:

Delete subinterfaces on Router.

Lecture 19: DTP/VTP

DTP — Dynamic Trunking Protocol. Automatically negotiates whether a switch port becomes a trunk. ISL is favored over dot1q.
Manual configuration is recommended for security.

DTP Modes

switchport mode dynamic desirable — actively tries to form a trunk.

switchport mode dynamic auto — passively waits for the other side.

switchport nonegotiate - Disable DTP

Administrative Mode	Trunk	Dynamic Desirable	Access	Dynamic Auto
Trunk	Trunk	Trunk	X	Trunk
Dynamic Desirable	Trunk	Trunk	Access	Trunk
Access	X	Access	Access	Access
Dynamic Auto	Trunk	Trunk	Access	Access

VTP — VLAN Trunking Protocol. Distributes and synchronizes VLAN (database) information between switches.
VTP does not create trunks.

DB-Sync-Requirements: Same VTP domain. A compatible VTP version. Password, when configured. A trunk connection.

VTP Modes:

Server: Can create, modify, and delete VLANs.

Stores the VLAN database.

Sends and receives VTP advertisements.

Can synchronize with a database that has a higher revision number.

Client: Cannot create, modify, or delete VLANs locally.

Receives VLAN information from VTP servers.

Forwards VTP advertisements.

Transparent: Does not synchronize its VLAN database.

VLANs are configured locally.

Can forward VTP advertisements.

Off: Does not synchronize VLANs.

Does not forward VTP advertisements.

Revision Number: Increases after VLAN database changes.

A higher revision number represents a newer VLAN database.

A switch with an incorrect high revision number can overwrite valid VLAN information in VTPv1 or VTPv2.

Check it before connecting a switch: **show vtp status**

Basic Configuration:

vtp domain CCNA

vtp mode [server/client]

vtp version 2

vtp password cisco

Versions: VTPv1/v2: Basic VLAN distribution.

VTPv3: Supports extended VLANs, better security, and primary-server operation.

show vtp versions .

Lecture 20-21:

STP (Spanning Tree Protocol)

Redundancy provides backup links but can cause Layer 2 loops (broadcast storms).

STP (IEEE 802.1D) prevents Layer 2 loops by blocking redundant paths.

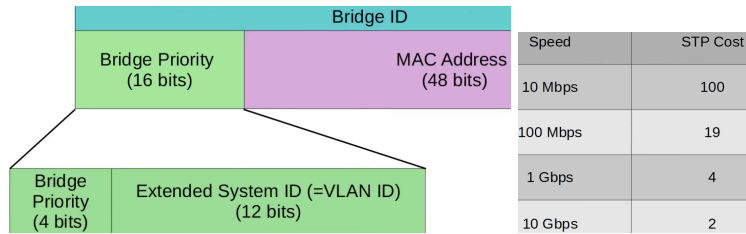
Forwarding ports carry traffic; blocking ports only process BPDUs to elect a root bridge and select port roles.

Root Port: Port of Non-Root-Switch to Root Bridge.

Designated Port: Forwarding Port on Switch-to-Switch-Segment.

Non-designated Port: Redundant (blocking) Port.

STP Process



#show spanning tree =>

STP Port ID = port priority (default 128) + port number

Interface	Role	Sts	Cost	Prio.Nbr	Type
G10/0	Desig	FWD	4	128.1	Shr
G10/1	Desig	FWD	4	128.2	Shr
G10/2	Desig	FWD	4	128.3	Shr
G10/3	Desig	FWD	4	128.4	Shr
G11/0	Desig	FWD	4	128.5	Shr
G11/1	Desig	FWD	4	128.6	Shr
G11/2	Desig	FWD	4	128.7	Shr
G11/3	Desig	FWD	4	128.8	Shr

1. Elect the root bridge: The switch with the lowest Bridge ID:

- Lowest priority
- Lowest MAC

All its ports are designated and forwarding.

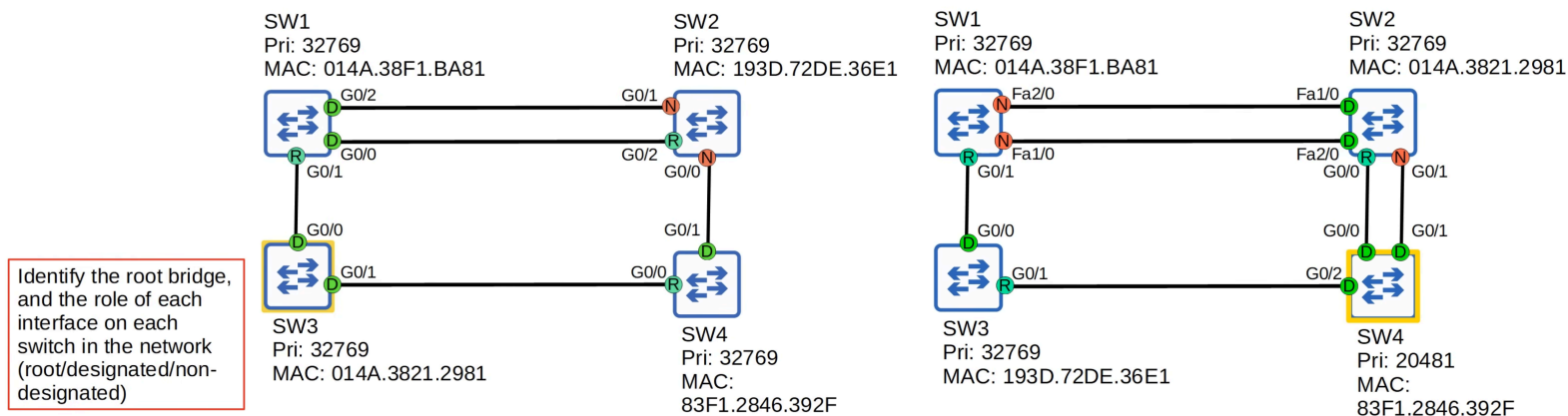
2. Select root ports: Each non-root switch selects one root port:

- Lowest root cost
- Lowest neighbor Bridge ID
- Lowest neighbor Port ID

3. Select designated ports: On each remaining link, the port on the switch with:

- Lowest root path cost
- Lowest Bridge ID

becomes the designated port. The other port blocks.



STP ports can have **5 different states**:

Listening: Only Root & Designated Ports can enter this state. Normally during 15 sec.

The port only sends and receives STP BPDUs. No normal network traffic sent/received. It does not learn MAC addresses.

Learning: It comes after the Listening state. Only Root & Designated Ports can enter this state. Normally during 15 sec.

The port only sends and receives STP BPDUs. No normal network traffic sent/received. It learns MAC addresses.

STP Port State	Send/Receive BPDUs	Frame forwarding (regular traffic)	MAC address learning	Stable/ Transitional
Blocking	NO/YES	NO	NO	Stable
Listening	YES/YES	NO	NO	Transitional
Learning	YES/YES	NO	YES	Transitional
Forwarding	YES/YES	YES	YES	Stable
Disabled	NO/NO	NO	NO	Stable

Disabled, when shutdown.

STP timers:

Hello timer – 2 seconds: The Root Bridge sends a BPDU every 2 seconds.

Forward Delay – 15 seconds: A port stays 15 seconds in Listening and 15 seconds in Learning before Forwarding.

Max Age – 20 seconds: If a switch does not receive BPDUs for 20 seconds, it assumes there is a network change and STP recalculates.

Only Root Bridge sends BPDUs. Other switches forward them to their designated ports.

STP Toolkit:

Portfast: Enabled on switch interfaces that are connected to endhosts. Immediately to Forwarding state, skip 30 sec wait.

(config-if)# **spanning-tree portfast**

(config)# **spanning-tree portfast default** enables on all interfaces for all access ports, but not for trunk ports.

BPDU-Guard: If the port receives a BPDU from another switch, it err-disables the port.

(config-if)# **spanning-tree bpduguard enable**

(config)# **spanning-tree portfast bpduguard default** enables on all interfaces.

To enable port again: **shutdown** => **no shutdown**.

Root Guard: Prevents a newly connected switch from becoming the root bridge. If a better BPDU comes in → port is blocked.

(config-if)# **spanning-tree guard root**

Loop Guard: Shuts down an interface if it stops receiving BPDUs. If no BPDU comes in anymore → port is blocked.

(config-if)# **spanning-tree guard loop**

SW(config)# **spanning-tree mode pvst** enable pvst

SW(config)# **spanning-tree vlan <vlan-id> root primary** make this sw to root bridge (lowest Bridge ID)

SW(config)# **spanning-tree vlan <vlan-id> root secondary** make this sw to backup root bridge (lowest Bridge ID)

SW#**show spanning-tree**

SW(config-if)# **spanning-tree vlan <vlan-id> cost <cost>** influences which STP path is preferred.

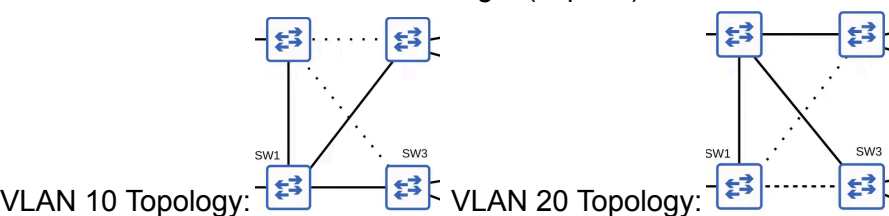
....**enable** => for 1

....**default** => for all

STP Load-Balancing: means different VLANs can use different paths. **e.g.**:

VLAN 10 uses SW1 as the Root Bridge.

VLAN 20 uses SW2 as the Root Bridge. (Top left)



Lecture 22:

Rapid STP (RSTP / 802.1w):

RSTP = faster version of STP.

Port states: Discarding, Learning, Forwarding.

Cisco commonly uses Rapid PVST+:

```
SW(config)# spanning-tree mode rapid-pvst enable rapid pvst
```

All Commands:

General:

? to view available commands.

> **enable** – enters **privileged EXEC mode**.

show startup-config – display the saved configuration.

show running-config – Displays the currently active configuration stored in RAM.

write / write memory / copy running-config startup-config –

configure terminal – enters **global configuration mode**.

(config)# **enable password <password>** – configure a case-sensitive password for privileged EXEC mode.

(config)# **service password-encryption** – encrypt plain-text passwords in the configuration.

(config)# **enable secret <password>** – configure a secure, hashed password for privileged EXEC mode.

(config)# **do <privileged EXEC command>** – executes a privileged EXEC command in a different mode.

(config)# **no <command>** – Removes or disables a configuration command.

MAC/IP:

show arp - view arp table: IP-Adresse → MAC-Adresse. Layer 3 zu Layer 2.

show mac address-table MAC-Adresse → Switchport/VLAN. Layer 2.

show ip interface brief – show a overview of all interfaces, IP addresses, and status values.

show interfaces [interface]

show interfaces description

(config)# **interface GigabitEthernet0/1** – Enters interface configuration mode for GigabitEthernet0/1.

(config-if)# **clear mac address-table dynamic [interface interface-id] / [address mac-id]**

(config-if)# **ip address ip subnet-mask** – Assigns the IP address to the interface.

(config-if)# **no shutdown** – activate the interface. **shutdown** – deactivate the interface.

show interfaces status

(config-if)# **duplex <value>** - set duplex-value

(config-if)# **speed <value>** - set speed-value

(config-if)# **description <value>** - set description-value

(config)# **interface range f0/2 - 3, f0/5 - 7** - to config interfaces 2,3,5,6,7 at the same time. **interface range config mode**.

Static Route Configuration:

R# **show ip route** - view routing table on routers.

R(config)# **ip route destination-ip netmask next-hop-ip**

R(config)# **ip route destination-ip netmask exit-interface** (Avoid, since it is directly connecting => Proxy ARP)

R(config)# **ip route destination-ip netmask exit-interface next-hop-ip**

e.g.: R(config)# **ip route 192.168.4.0 255.255.255.0 g0/0 192.168.24.4**

VLAN:

#**show vlan brief**

Configure Switch:

SW(config-if)#**switchport mode access** - Configures the selected switch ports as access ports.

SW(config-if)#**switchport access vlan <vlan-id>** - Configure the VLAN of a switch access port.

SW(config)#**vlan <vlan-id>** - new vlan

SW(config-vlan)#**name <name>** - name vlan

SW#**show interfaces trunk** - Displays the configured trunk ports and their VLAN information.

SW(config-if)#**switchport trunk encapsulation dot1q** - Configures the trunk port to use IEEE 802.1Q encapsulation.

SW(config-if)#**switchport mode trunk** - Configures the selected switch ports as trunk ports.

SW(config-if)#**switchport trunk allowed vlan <vlan-id>** - specifies which VLANs are allowed to pass through a trunk port.

SW(config-if)#**switchport trunk native vlan <vlan-id>** - Changes the native VLAN of the trunk port.

Configure Router:

R(config-if)#**interface interface-id <vlan-id>** e.g.: **interface g0/0.10** - Creates and enters a router s

R(config-subif)#**encapsulation dot1q <vlan-id> native** - Assigns native VLAN.ubinterface.

R(config-subif)#**encapsulation dot1q <vlan-id>** - Assigns the subinterface to a VLAN.

R(config-subif)#**ip address ip netzmask** - Assigns an IP address and subnet mask to subinterface. Address = default gw.

Configure L3-Switches:

SW(config)#**ip routing** - enables Layer 3 routing on the switch.

(config-if)#**no switchport** - configures the interface as 'routed' (L3 port) **interface vlan10** => assign Ip-Address => **no shutdo**

default interface g0/1 - to default settings.

DTP/VTP:

#**show vtp status**

#**show vtp versions**

DTP Modes

(config-if)#**switchport mode dynamic desirable** — actively tries to form a trunk.

(config-if)#**switchport mode dynamic auto** — passively waits for the other side.

(config-if)#**switchport nonegotiate** - Disable DTP

(config-if)#**sw mode access/trunk** - Disable DTP

VTP Configuration:

(config)#**vtp domain CCNA**

(config)#**vtp mode [server/client/transparent]**

(config)#**vtp version 2**

(config)#**vtp password cisco**

Which IP-address to enter:

ip address → IP of the local interface you are configuring

ip route → destination network + next-hop router interface IP

ping/SSH/Telnet → IP of the destination device

STP Toolkit:

Portfast:

(config-if)# **spanning-tree portfast**

(config)# **spanning-tree portfast default** enables on all interfaces for all access ports, but not for trunk ports.

BPDU-Guard:

(config-if)# **spanning-tree bpduguard enable**

(config)# **spanning-tree portfast bpduguard default** enables on all interfaces.

To enable port again: **shutdown** => **no shutdown**.

Root Guard:

(config-if)# **spanning-tree guard root**

Loop Guard:

(config-if)# **spanning-tree guard loop**

SW(config)# **spanning-tree mode pvst** enable pvst

SW(config)# **spanning-tree vlan <vlan-id> root primary** make this sw to root bridge

SW(config)# **spanning-tree vlan <vlan-id> root secondary** make this sw to backup root bridge

SW#**show spanning-tree**

SW(config.if)# **spanning-tree vlan <vlan-id> cost <cost>** influences which STP path is preferred.

Ultra short recap:

Subnetting: Splits an IP network into smaller subnets to improve address usage.

VLAN: Splits a Layer 2 network into separate broadcast domains.

DTP = Negotiates whether a switch port becomes a trunk.

VTP = Distribute and synchronize VLAN information between switches.

STP = Prevents Layer 2 switching loops by blocking redundant network paths.